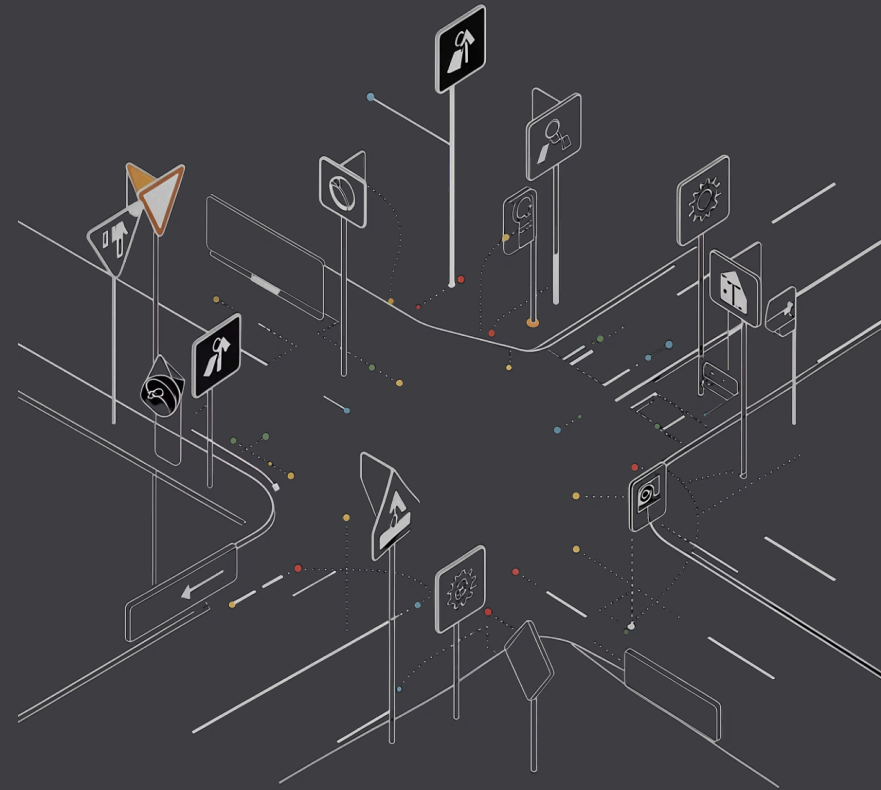


Two-stage road sign detection system

YOLO + 5xResNet-18 | WebSocket | Live HTTPS demo



Problem: critical class imbalance



55 classes — distribution is extremely uneven

Class 22: **327 samples**, class 7: **only 2**. Imbalance ratio: **163:1**.

With such skew, standard accuracy is misleading. **Macro Recall** is the only metrically honest indicator for rare signs.

⚠️ Rare classes should not «sink» into background categories — this is the key limitation of the task.

Data-Level Solutions

Oversampling

×10–×20 for critical classes (<10 samples), ×2–×5 for rare ones (10–50 samples). Forced distribution balancing.

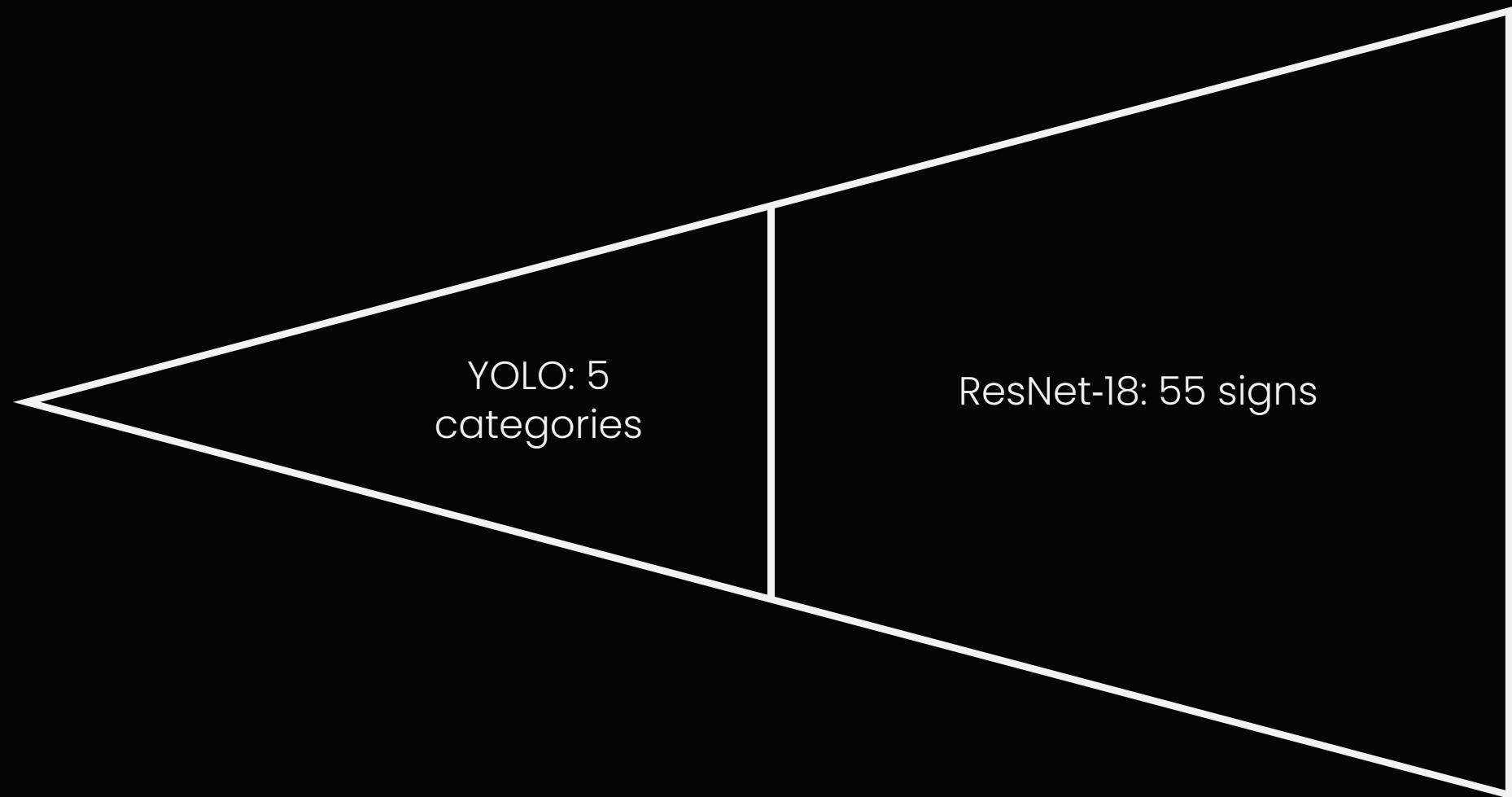
Strong augmentations

RandomRotate90, ShiftScaleRotate, RandomBrightnessContrast, GaussNoise, Blur — simulation of real shooting conditions.

Result

Recall **98–100%** on prohibitory, warning, informational, and priority signs.

Architecture: two-stage pipeline



YOLO works as a detector and coarse classifier, while ResNet-18 is a precise refinement classifier.

Pure YOLO (55 classes)

Recall: **82%** — direct training on all classes does not handle rare signs well.

Two-stage approach

Recall: **91.7%** — splitting the task into detection and refinement gives a gain of +9.7 pp.

ResNet-18 Training Details

Initialization

Pre-training on ImageNet, full fine-tuning of all layers.

Loss function

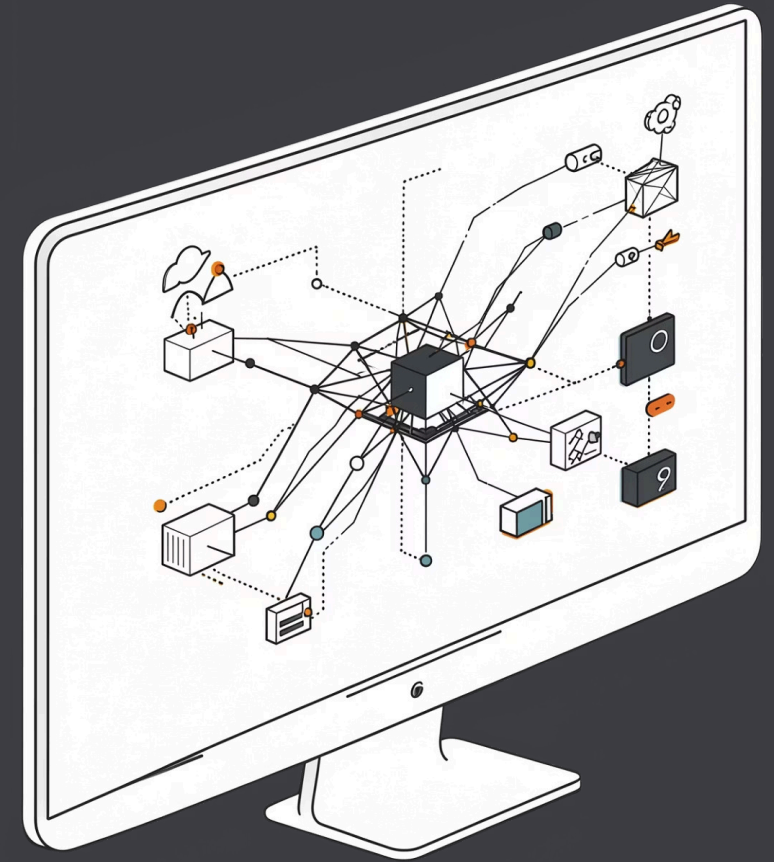
CrossEntropy + label smoothing (0.1) — regularization against overfitting on small classes.

Optimization

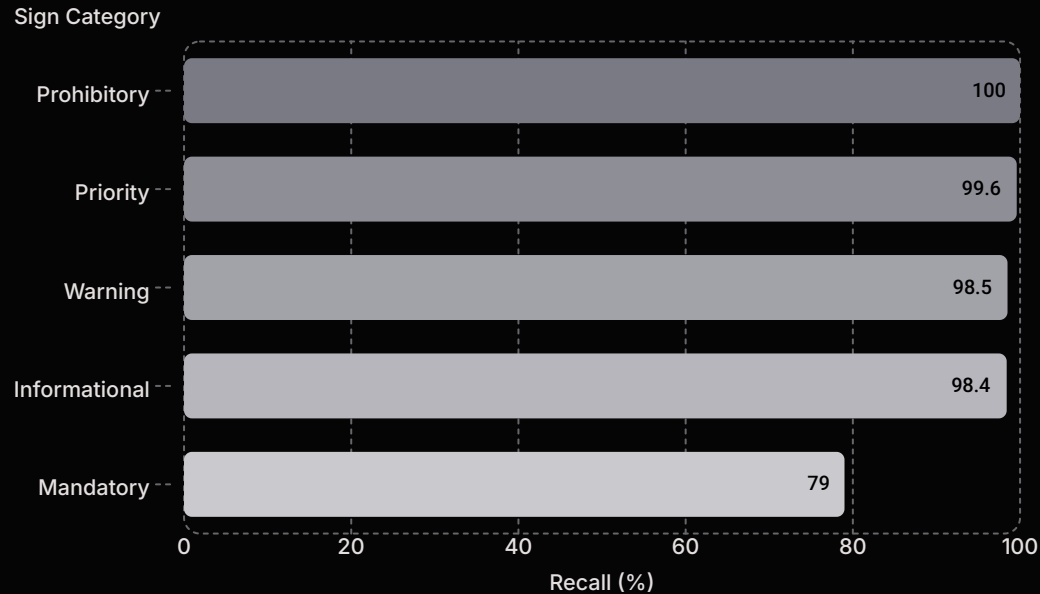
AdamW + ReduceLROnPlateau — adaptive LR responds to validation plateaus.

Sampling

Positives: YOLO crops with $\text{IoU} \geq 0.5$. Background: false positives, missed signs, random crops (10%).



Results by Category



Strengths

Our model demonstrates strong performance for prohibitory, priority, warning, and informational signs, consistently achieving 98-100% recall rates. This ensures critical road information is reliably detected.

Weak Spot

Mandatory signs currently stand at 79% recall. This is primarily due to visual ambiguity arising from similar arrow directions on different mandatory signs, making differentiation challenging for the model.

 Our target threshold for mandatory signs is 90% recall.

System Metrics on the Test Set

94.1%

mAP50

Detection accuracy at IoU ≥ 0.5

85.6%

mAP50–95

Strict mAP across the IoU range

90.3%

Precision

Share of correct detections

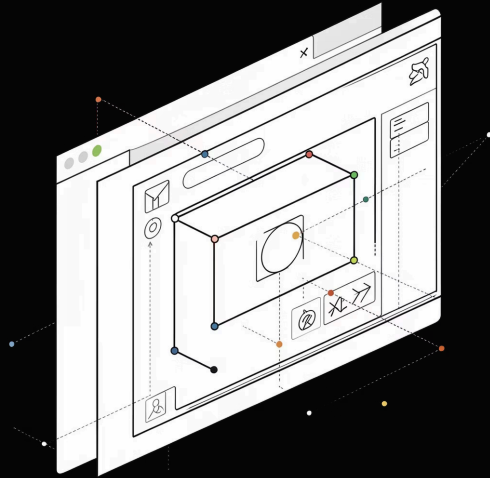
91.7%

Recall

Detection completeness for signs

The hybrid architecture outperforms pure YOLO by **+12 pp in recall** and **+7.8 pp in precision** — a significant gain from two-stage classification.

Real-time inference and deployment



Inference pipeline

01

The browser sends a frame via **WebSocket** every 300 ms

02

Server: YOLO detects → crop → ResNet classifies

03

JSON response: coordinates, class, confidence

04

The client draws **green boxes** with sign names

Deployment: **Docker + Caddy HTTPS + WebSocket**

Demo and Code

Live demo

Open <https://greencat1.tech> in a browser and allow camera access.
Detection runs in real time via WebSocket.

GitHub repository

Full project code:

github.com/greencat1/Portfolio/tree/main/pet_projects/traffic-sign-detection — training, inference, deployment.



Conclusions and Outlook

✓ Achieved

Recall **91.7%**,
Precision **90.3%** —
the hybrid
architecture
significantly
outperforms single-
stage YOLO (+12 pp
recall).

⚠ Area for Improvement

Prescriptive signs:
current recall **79%**,
target — **90%**.
Architecture or data
improvements are
needed.

➡ SOON Next Step

LLM assistant for
working with sign
documentation — on
the same
deployment stack
(Docker +
WebSocket).

